VENDORGATE

Vendor Onboarding and Purchase Order Portal

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

Case Study No. 88

Document Attribute	Details
Document Type	Software Requirements Specification (SRS)
Project	VendorGate – Vendor Onboarding and Purchase Order Portal
Case Study	Case Study No. 88
Organization Context	Mid-size pharmaceutical distributor
Vendor Base	850 vendors
Required FRs	12
Required NFRs	8
SRS Style	IEEE 830-style academic specification
Version	1.0
Status	Draft / Academic Project Submission

1. Introduction

1.1 Purpose

This Software Requirements Specification defines the software requirements for VendorGate, a Vendor Onboarding and Purchase Order Portal. It translates the business baseline established in the BRD and SOW into detailed, testable software requirements.

1.2 Scope

VendorGate will provide a centralized process for vendor registration, document submission and verification, purchase-order creation, approval-limit routing, purchase-order approval, vendor acknowledgement and auditability.

1.3 Intended Audience

- Project Manager
- Business Analyst
- Solution / Software Design Team
- Development Team
- Testing / QA Team
- Procurement stakeholders
- Finance stakeholders
- Audit stakeholders
- Vendor representatives

2. Product Overview

The organization currently works with 850 vendors. Vendor registration involves 11 documents sent by email, while purchase orders are raised in Excel and approved on paper. The audit team found 34 purchase orders approved above the approver's limit last year.

VendorGate addresses these problems by providing one controlled digital workflow for vendor onboarding and purchase-order approval.

3. User Groups / Business Actors

Actor	Description	Primary Activities
Vendor	External participant in the portal.	Register, submit documents, view PO information, acknowledge PO.
Procurement User	Operational business user.	Manage vendor onboarding and create/manage purchase orders.
Approver	Authorized internal user with an applicable approval level.	Review and approve/reject purchase orders within authority.
Finance	Financial control stakeholder.	Define/maintain approval-limit policy and participate in financial approval governance.
Audit	Independent control stakeholder.	Review traceability, approval records and control evidence.

System Administrator	System administration role for controlled access and configuration.	Manage users/roles and system-level administration as authorized.

4. System Scope

4.1 In Scope

- Vendor registration and vendor information capture.
- Submission and verification of required vendor documentation.
- Purchase-order creation and management.
- Approval-limit based routing and authorization control.
- Vendor acknowledgement of purchase orders.
- Auditability and traceability.
- Role-based access appropriate to user responsibilities.

4.2 Out of Scope

- Payment processing.
- Inventory management.
- Invoice management.
- Shipment tracking.
- Accounting or ERP integration.
- Mobile application.
- AI-based vendor assessment.
- Other features not specified by the case study unless approved through change control.

5. System Actors and Access Principles

The system shall enforce role-appropriate access. Users shall only perform actions permitted by their role. The exact production identity provider or technology implementation is intentionally not specified by the case study.

6. Functional Requirements

Exactly 12 functional requirements are specified below, matching the quantity required by the case study. Priorities use MoSCoW.

ID	Feature	MoSCoW	Requirement	Rationale	Precondition	Expected Behavior	Acceptance Ref
FR-01	Vendor Registration	Must	The system shall allow a vendor to create and submit a registration record containing the information required by the approved	Vendor registration is the primary onboarding capability.	Valid registration information is available; vendor is permitted to register.	Registration submitted successfully or validation errors are shown.	AC-FR01

			vendor-registration process.				
FR-02	Vendor Document Submission	Must	The system shall allow a vendor to submit the required registration documents through the portal.	The existing process requires 11 documents to be exchanged through email.	Vendor registration record exists.	Documents are attached to the correct registration and their submission status is recorded.	AC-FR02
FR-03	Document Verification	Must	The system shall allow an authorized internal user to review submitted vendor documents and record the verification outcome.	Document verification is explicitly required by the case.	Documents have been submitted.	Each document/review outcome is recorded and the vendor registration status reflects the verification outcome.	AC-FR03
FR-04	Vendor Status Management	Must	The system shall maintain the vendor registration status through the defined onboarding process.	A centralized onboarding workflow requires a visible registration state.	Vendor registration exists.	Authorized users can view the current vendor status and the status changes are traceable.	AC-FR04
FR-05	Purchase Order Creation	Must	The system shall allow an authorized Procurement user to create a purchase order for a registered vendor.	The current PO process is Excel-based and the proposed portal requires PO creation.	An eligible vendor exists and the Procurement user is authorized.	A PO is created with a unique reference and required PO information.	AC-FR05
FR-06	Approval-Limit Determination	Must	The system shall determine the applicable approval level for a purchase order using the four approval limits specified in the case study.	Approval-limit control is required to address the audit finding.	PO amount is available.	The system assigns the correct approval level according to the defined boundaries.	AC-FR06
FR-07	PO Approval Routing	Must	The system shall route a purchase order to an approver	The case requires PO creation with approval limits and	Approval level has been determined.	The PO is routed to the applicable approval authority and	AC-FR07

			authorized for the determined approval level.	controlled routing.		cannot bypass the defined route through the normal workflow.	
FR-08	PO Approval Decision	Must	The system shall allow an authorized approver to approve or reject a purchase order and record the decision.	A digital approval workflow replaces paper approval.	PO has been routed to the approver.	The decision, status and relevant audit information are recorded.	AC-FR08
FR-09	Vendor PO Acknowledgement	Must	The system shall allow the vendor to view an applicable purchase order and record an acknowledgement decision.	Vendor acknowledgement is explicitly required by the case.	PO is available to the vendor.	The acknowledgement status is recorded against the PO.	AC-FR09
FR-10	Role-Based Access Control	Must	The system shall restrict actions according to the user's assigned role and authorization.	Procurement, Finance, Audit, vendors and approvers have different responsibilities.	User identity and role are known.	Unauthorized actions are blocked and authorized actions are available.	AC-FR10
FR-11	Audit Trail	Must	The system shall maintain traceable records of relevant vendor-registration, document-verification, PO, approval and acknowledgement activities.	Auditability is explicitly required by the case and the audit finding makes traceability important.	A tracked business action occurs.	The relevant action, actor, time and status information is recorded and retrievable by authorized users.	AC-FR11
FR-12	Dashboard / Tracking	Should	The system shall provide authorized users with status information for vendor registrations and purchase orders relevant to their role.	Centralized tracking supports the business need for visibility over onboarding and PO workflows.	Records exist for the user's scope.	The user can view relevant current statuses and identify pending/complete workflow items.	AC-FR12

7. Non-Functional Requirements

The case study requires exactly 8 non-functional requirements and states that each must have a measurable target, particularly for auditability, security and availability. The numerical targets below are explicitly marked as project-defined targets because the case study does not supply those numbers.

ID	Category	MoSCoW	Measurable Requirement	Reason / Basis	Evidence
NFR-01	Availability	Must	Project-defined target: VendorGate shall achieve at least 99.0% availability during the agreed operating window, measured monthly.	Availability is explicitly required by the case study.	Availability monitoring records.
NFR-02	Performance	Should	Project-defined target: 95% of standard portal requests shall receive a response within 3 seconds under the defined test load.	Provides a measurable performance target without claiming the case supplied it.	Performance test results.
NFR-03	Security / Access Control	Must	100% of protected functional actions shall require an authenticated and authorized role; unauthorized access attempts shall be denied.	Security is explicitly required.	Security and authorization test results.
NFR-04	Auditability	Must	100% of defined auditable business events (registration submission, document verification, PO creation, approval/rejection and acknowledgement) shall create an audit record.	Auditability is explicitly required.	Audit-event completeness test.
NFR-05	Audit Record Integrity	Must	100% of audit records shall contain, at minimum, actor, event type, timestamp and related business-record reference.	Makes audit evidence measurable.	Audit-record validation test.

NFR-06	Data Integrity	Must	100% of approval-routing test cases shall return the correct approval level for the defined boundary and equivalence-class inputs.	Approval routing is a core control in the case.	Decision-table/BVA test results.
NFR-07	Usability	Should	Project-defined target: at least 90% of representative task-based usability test participants shall complete the defined core task flow without assistance.	Makes usability measurable without inventing a case-study user count.	Usability test evidence.
NFR-08	Maintainability / Testability	Should	Project-defined target: 100% of functional requirements shall have at least one traceable test case in the RTM before final acceptance.	The case explicitly requires traceability from every FR to a test case.	Completed RTM review.

8. Technology Stack & Implementation Environment

The following is the proposed implementation stack for the academic VendorGate prototype. These technology choices are implementation decisions and are not case-study-supplied requirements.

Area	Technology / Choice	Purpose / Description
Frontend	React.js.	VendorGate web interface.
Backend	Node.js	Server-side runtime.
Server Framework	Express.js	REST API and backend routes.
API Style	REST API	Frontend-backend communication.
Data	Seed / Mock Data	Predefined prototype data.
Database	No persistent database in academic prototype	A persistent database can be added later.
Version Control	Git / GitHub	Source-code versioning.
Development Environment	VS Code / Antigravity IDE	Development environment.
UI	Custom Responsive Web Interface	Responsive application interface.

9. Business Rules

Rule ID	Business Rule
BRULE-01	Vendor registration requires the existing set of 11 registration documents; the case study does not name the documents individually.
BRULE-02	Purchase orders must use the four approval levels supplied by the case study.
BRULE-03	Level 1 applies to amounts up to ₹50,000.
BRULE-04	Level 2 applies to amounts above ₹50,000 and up to ₹2,00,000.
BRULE-05	Level 3 applies to amounts above ₹2,00,000 and up to ₹10,00,000.
BRULE-06	Level 4 applies to amounts above ₹10,00,000.
BRULE-07	An approver must not approve a PO beyond the authority represented by the applicable approval level.
BRULE-08	Vendor acknowledgement is performed after the applicable PO approval workflow.

10. Approval Policy and Boundary Definitions

Boundary	Below / Lower Class	Boundary Value	Above / Upper Class
₹50,000	₹49,999	₹50,000	₹50,001
₹2,00,000	₹1,99,999	₹2,00,000	₹2,00,001
₹10,00,000	₹9,99,999	₹10,00,000	₹10,00,001

These values are derived from the approval limits explicitly supplied by the case study and are included here as the basis for Boundary Value Analysis. The detailed test cases and decision table will appear in the Test Plan.

11. Vendor Registration Requirements

- A vendor shall be able to initiate registration through VendorGate.
- The registration process shall support submission of the required 11 documents.
- The system shall associate submitted documents with the correct vendor registration.
- Authorized users shall be able to review and verify submitted documents.
- The registration status shall be visible to authorized users.
- The system shall retain traceable registration and verification activity.

12. Purchase Order Requirements

- Authorized Procurement users shall be able to create POs for eligible vendors.
- The PO amount shall be used to determine the applicable approval level.
- The PO shall be routed to an authorized approver.
- The approval decision shall be recorded.
- The system shall prevent normal workflow approval by an authority whose applicable limit is insufficient.
- The PO status shall be updated following the approval decision.
- The PO shall be available for vendor acknowledgement after the relevant approval stage.

13. Vendor Acknowledgement Requirements

- An applicable vendor shall be able to view the PO presented for acknowledgement.
- The vendor shall be able to record an acknowledgement decision.
- The acknowledgement status shall be associated with the PO.
- The acknowledgement activity shall be included in the audit trail.

14. Security and Audit Requirements

- Users shall be authenticated before accessing protected functions.
- Role-based authorization shall be enforced.
- Vendor users shall not access other vendors' restricted information through normal system functions.
- Approvers shall only perform approval actions within their authorized level.
- Relevant business actions shall be recorded for auditability.
- Audit records shall be protected from unauthorized modification through appropriate access controls.

15. Constraints

- The project must use the quantitative figures provided by Case Study No. 88 and must not replace them with invented figures.
- The four approval limits supplied in the case must be used.
- The case requires exactly 12 functional requirements and 8 non-functional requirements in the SRS.
- The 11 vendor registration documents are referenced as a count only; their individual names are not supplied by the case study.
- The SRS must support the UML, testing, scheduling, stakeholder and risk deliverables required by the case.

16. Assumptions

- Vendors will use VendorGate for the registration and acknowledgement activities within the project scope.
- Procurement users will be authorized to create and manage POs.
- Authorized approvers will have defined approval authority corresponding to the four levels.
- Finance and Audit will participate in governance and requirements validation.
- The approval-limit policy will be confirmed before the PO approval workflow is treated as finalized.

17. Dependencies

Activity	Duration	Dependency
Requirements	8 days	None
Approval-limit policy sign-off	6 days	After Requirements
Vendor registration module	14 days	After Requirements
PO module	16 days	After Approval-limit policy sign-off
Vendor acknowledgement	6 days	After PO module
Testing	8 days	After Vendor registration and Vendor acknowledgement
Vendor onboarding drive	15 days	After Testing

18. SRS Acceptance Criteria

ID	Acceptance Criterion
SRS-AC01	All 12 functional requirements are uniquely identified and testable.
SRS-AC02	All 8 NFRs have measurable targets and their targets are identified as project-defined where not supplied by the case.
SRS-AC03	The four approval limits are represented correctly.
SRS-AC04	Vendor registration, document verification, PO creation, approval, acknowledgement and auditability are covered.
SRS-AC05	MoSCoW priorities are assigned to the requirements.
SRS-AC06	Requirements are traceable to test cases through the RTM.
SRS-AC07	No unsupported case-study figures or names of the 11 documents are presented as facts.

19. Requirements Traceability Matrix (RTM)

The case study requires every functional requirement to be linked to a test case. The test-case IDs below are the planned traceability references and will be expanded with full test evidence in the Test Plan.

FR ID	Requirement	Test Case ID	Test Focus
FR-01	Vendor Registration	TC-01	Vendor registration submission
FR-02	Vendor Document Submission	TC-02	11-document submission / completeness validation
FR-03	Document Verification	TC-03	Authorized verification outcome
FR-04	Vendor Status Management	TC-04	Registration status transition
FR-05	Purchase Order Creation	TC-05	Valid PO creation
FR-06	Approval-Limit Determination	TC-06, TC-07, TC-08	Boundary and equivalence routing
FR-07	PO Approval Routing	TC-09	Correct approver routing / bypass control
FR-08	PO Approval Decision	TC-10	Approve/reject decision
FR-09	Vendor PO Acknowledgement	TC-11	Vendor acknowledgement
FR-10	Role-Based Access Control	TC-12	Unauthorized/authorized action checks
FR-11	Audit Trail	TC-13	Audit-event completeness and traceability
FR-12	Dashboard / Tracking	TC-14	Role-relevant status tracking

20. MoSCoW Summary

Priority	Requirements
Must Have	FR-01 to FR-11; NFR-01, NFR-03, NFR-04, NFR-05, NFR-06
Should Have	FR-12; NFR-02, NFR-07, NFR-08
Could Have	No additional case-supported requirements are added to the baseline.
Won't Have – Current Baseline	Features outside the case scope such as payments, inventory, invoices, shipment tracking and ERP integration.

21. Data and Interface Requirements

The case study requires a software portal but does not specify a particular database, API technology or external integration. Therefore, this SRS defines logical data requirements rather than inventing a technology stack.

Logical Data Area	Required Information
Vendor	Registration information and current status.
Vendor Document	Document associated with a vendor registration and its verification outcome.
Purchase Order	PO reference, vendor association, amount, status and approval information.
Approval	Approval level, approver, decision and decision status.
Acknowledgement	Vendor acknowledgement decision and related PO.
Audit Record	Actor, event, timestamp, related record and outcome.
User / Role	User identity and assigned role/authorization.

22. Validation and Error Handling

- Required registration information shall be validated before submission.
- Required document submission conditions shall be checked before a registration is accepted as complete.
- A PO shall not proceed through approval without a valid approval-level determination.
- Unauthorized approval actions shall be rejected.
- Invalid or incomplete user actions shall result in clear validation feedback.
- Failed workflow actions shall not silently appear as successful transactions.

23. Related Project Deliverables

Deliverable	Relationship to this SRS
UML Package	Uses the actors, requirements and workflows defined here.
Project Plan	Uses the project activities and dependencies supplied by the case.
Stakeholder & RACI	Defines governance and responsibilities around the requirements.
Test Plan & Evidence	Tests the FRs, approval boundaries, decision table and NFR targets.
Risk Register & Closure Note	Manages the four risks supplied by the case and project responses.

24. Terminology

Term	Meaning
FR	Functional Requirement – what the system must do.
NFR	Non-Functional Requirement – measurable quality or constraint on how the system operates.
MoSCoW	Must Have, Should Have, Could Have, Won't Have prioritization.
PO	Purchase Order.
Approval Limit	Maximum PO amount that can be approved at a given authority level.
BVA	Boundary Value Analysis – testing values around limits.
RTM	Requirements Traceability Matrix – maps requirements to test cases.
Audit Trail	Traceable record of relevant system/business activities.

25. Conclusion

This SRS converts the VendorGate business baseline into a detailed and testable software specification. It contains exactly 12 functional requirements and 8 measurable non-functional requirements as required by the case study. The four approval levels and their boundaries are explicitly represented, while unsupported information such as the names of the 11 registration documents has not been invented.

The SRS is intended to serve as the baseline for the UML Package, Project Plan, Test Plan and Requirements Traceability Matrix, and later for implementation of the VendorGate prototype.